# MTH633
## SOLVED QUESTIONS

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**Q.No.1:** Define commutator subgroup. **Marks = 02**

**ANSWER:** The set of all commutators $aba^{-1}b^{-1}$ for $a, b \in G$ generates a subgroup $C$ of $G$ is called the commutator subgroup.

**Q.No.2:** State Cauchy's Theorem. **Marks = 02**

**ANSWER:** Let $p$ be a prime. Let $G$ be a finite group and let $p$ divide $|G|$. Then $G$ has an element of order $p$ and, consequently, a subgroup of order $p$.

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**Q.No.3:** Is $(\mathbb{Z}_n, +)$ a cyclic group? If yes then tell its generator.

Marks = 02

**ANSWER:** $(Zn, +)$ is a cyclic group with $1$ as a generator. When there is no confusion, we write the elements of $Z_n$ as $0, 1, 2, 3, \ldots, n-1$ instead of $[0], [1], [2], [3], \ldots, [n-1]$.

**Q.No.4:** Define Simple Groups. Marks = 02

**ANSWER:** A group is simple if it is nontrivial and has no proper nontrivial normal subgroups.

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**Q.No.5:** Show that $Z(G)$ is a normal and an abelian subgroup of G.

Marks = 3

**ANSWER:** For each $g \in G$ and $z \in Z(G)$ we have

$$gzg^{-1} = zgg^{-1} = ze = z,$$

we see at once that $Z(G)$ is a normal subgroup of $G$.

It implies that

$$gz = zg \text{ for } g \in G \text{ and } z \in Z(G).$$

If $G$ is abelian, then

$$Z(G) = G;$$

In this case, the center is not useful.

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**Q.No.6:** Show that a factor group of a cyclic group is cyclic.

Marks = 3

**ANSWER:** Let $G$ be cyclic with generator $a$, and let $N$ be a normal subgroup of $G$.

We claim the coset $aN$ generates $G/N$. We must compute all powers of $aN$.

But this amounts to computing, in $G$, all powers of the representative $a$ and all these powers give all elements in $G$.

Hence the powers of $aN$ certainly give all cosets of $N$ and $G/N$ is cyclic.

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**Q.No.7:** Let G be a group. The set of all commutators $aba^{-1}b^{-1}$ for $a, b \in G$ generates a subgroup C of G. Marks = 3

**ANSWER:** Let $a, b \in G$.

Then, $(aba^{-1}b^{-1})(aba^{-1}b^{-1})^{-1}$

$= aba^{-1}b^{-1}bab^{-1}a^{-1}$

$= e \in C$

since $e = eee^{-1}e^{-1}$ is a commutator.

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**Q.No.8:** If N is a normal subgroup of G, then show that G/N is abelian if and only if $C \leq N$, where $C$ is commutator subgroup.

Marks = 3

**ANSWER:** If N is a normal subgroup of G and G/N is abelian, then

$$(a^{-1}N)(b^{-1}N) = (b^{-1}N)(a^{-1}N);$$

that is, $aba^{-1}b^{-1}N = N$,

so $aba^{-1}b^{-1} \in N$, and $C \leq N$.

Finally, if $C \leq N$, then $(aN)(bN) = abN$

$= ab(b^{-1}a^{-1}ba)N$

$= (abb^{-1}a^{-1})baN$

$= baN = (bN)(aN).$

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**Q.No.9:** Let H be a subgroup of a group G. Then H is normal if and only if $(aH)(bH) = (ab)H$, for all $a, b$ in $G$. **Marks = 5**

**ANSWER:** Suppose $(aH)(bH) = (ab)H$, for all $a, b$ in $G$.

We show that $aH = Ha$, for all $a$ in $H$.

We do this by showing:

$aH \subseteq Ha$ and $Ha \subseteq aH$, for all $a$ in $G$.

$aH \subseteq Ha$: First observe that

$$aHa^{-1} \subseteq (aH)(a^{-1}H) = (aa^{-1})H = H.$$

Let $x$ be in $aH$. Then $x = a\,h$, for some $h$ in $H$. Then $xa^{-1} = aha^{-1}$, which is in $= aHa^{-1}$, thus in $H$. Thus $xa^{-1}$ is in $H$. Thus $x$ is in $Ha$.

$Ha \subseteq aH$: $Ha \subseteq HaH = (eH)(aH) = (ea)H = aH$.

This establishes nor...

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Let $x$ be in $aH$. Then $x = ah$, for some $h$ in $H$. Then $xa^{-1} = aha^{-1}$, which is in $= aHa^{-1}$, thus in $H$. Thus $xa^{-1}$ is in $H$. Thus $x$ is in $Ha$.

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$Ha \subseteq aH$: $Ha \subseteq HaH = (eH)(aH) = (ea)H = aH$.

This establishes normality.

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**Q.No.10:** Let $\varphi$ be a homomorphism function from group $(G,*)$ to group $(G',.)$. Then, $(Ker\ \varphi,*)$ is a normal subgroup of $(G,*)$. **Marks = 5**

Proof: i) $Ker\varphi$ is a subgroup of G

$\forall a, b \in Ker\varphi, \varphi(a) = e_{G'}, \varphi(b) = e_{G'}$.

Then, $(a * b) = \varphi(a)\ \varphi(b) = e_{G'}$. Therefore, $a * b \in Ker\varphi$.

Inverse element:

$\forall a \in Ker\varphi, \varphi(a) = e_{G'}$. Then, $\varphi(a^{-1}) = \varphi(a)^{-1} = e_{G'}$

Therefore, $a^{-1} \in Ker\varphi$.

ii) $\forall g \in G, a \in Ker\varphi, \varphi(a) = e_{G'}$. Then, $\varphi(g^{-1} * a * g)$

$= \varphi(g^{-1})\ \varphi(a)\varphi\ (g)$

$= \varphi\ (g)^{-1}\ e_{G'}\ \varphi\ (g)$

$= e_{G'}$ Therefore, $g^{-1} * a * g \in Ker\varphi$.

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**Q.No.11:** Let H and K be normal subgroups of a group G with $K \leq H$. Then show that $G/H \simeq (G/K)/(H/K)$ Marks = 5

**Proof:** Let $\phi: G \to (G/K)/(H/K)$ be given by $\phi(a) = (aK)(H/K)$ for $a \in G$.

Clearly $\phi$ is onto $(G/K)/(H/K)$, and for $a, b \in G$,

$\phi(ab) = [(ab)K](H/K)$

$= [(aK)(bK)](H/K)$

$= [(aK)(H/K)][(bK)(H/K)] = \phi(a)(b)$,

so $\phi$ is a homomorphism.

The kernel consists of those $x \in G$ such that $\phi(x) = H/K$.

These $x$ are just tha

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Clearly $\phi$ is onto $(G/K)/(H/K)$, and for $a, b \in G$,

$$\phi(ab) = [(ab)K](H/K)$$
$$= [(aK)(bK)](H/K)$$
$$= [(aK)(H/K)][(bK)(H/K)] = \phi(a)(b),$$

so $\phi$ is a homomorphism.

The kernel consists of those $x \in G$ such that $\phi(x) = H/K$.

These $x$ are just the elements of $H$. Then first isomorphism theorem shows that

$$G/H \simeq (G/K)/(H/K).$$

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**Q.No.12:** Show that the homomorphism $h: G \to G'$ is injective if and only if $Ker\ h = \{e\}$. Marks = 5

Proof: Suppose $h$ is injective, and let $x \in Ker\ h$.

Then $h(x) = e' = h(e)$. Hence $x = e$.

Conversely, suppose $Ker\ h = \{e\}$.

Then $h(x) = h(y)$

$\Rightarrow h(xy^{-1}) = h(x)h(y^{-1})$

$= h(x)h(y)^{-1} = e'$

$\Rightarrow xy^{-1} \in Ker\ h$

$\Rightarrow xy^{-1} = e$

$\Rightarrow x = y$.

Hence, $h$ is injective.

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